

CSCS 495 A

FORMAN CHRISTIAN COLLEGE
(A CHARTERED UNIVERSITY)
CYBER SECURITY
Assignment 3

Max Marks: 50

Submission Format:

In the report you should provide the following:

- Report should carry details about the scans and payload generation augmented with clear screen shots.
- In order to show that it is your work, create a user on kali with your initial name, reboot kali and login as this new user. Now carry out all steps while taking screen shots where the user name is prominently shown on kali prompt.
- A soft copy of the report should be uploaded on MOODLE on or before 11:59 pm, **Nov 24, 2020**. A hard copy of the reports should be submitted in class whenever we start with in person class room teaching.
- Make sure to clearly specify the Roll Number and Names of the group members on your reports. Two reports (one for each task should be prepared, zipped together and named with your roll number's last three digits should be provided.
- **Note that maximum two students in a group are allowed.**
- Note: Groups who have been working with three or four members can split into two groups and can share task 1 only. Task 2 must be completed by each group.
- Groups who are already working with two group members must not add additional member/s.

Task 1 [25]

In this task we will install OWASP BWA on our machines.

Detailed steps are given.

You should prepare a report carrying all these steps and appropriate screen shots (of your terminal/system screen) augmenting each step.

OWASP-BWA typically refers to the **OWASP Broken Web Applications** project, a set of intentionally vulnerable web applications used for security testing and educational purposes.

The project provides a collection of different vulnerable web applications that are designed to demonstrate a variety of common web application security issues, including but not limited to:

- Cross-site scripting (XSS)
- SQL injection
- Insecure cryptographic storage
- Cross-site request forgery (CSRF)
- Remote code execution (RCE)
- Directory traversal

These applications are often used by security professionals, penetration testers, and students learning about web application security to practice detecting and mitigating vulnerabilities in a controlled environment.

Installation steps to follow;

- Go to Google and type “owaspbwa download”
- Click on OWASP Broken Web Application Project Source Forge link
- Download the app on your kali machine.
- Extract it and save it on /home/kali/Documents/
- In the extracted folder you will see some VMs.
- Use OWASP Broken Wen Apps-cl1
- Note that its size is 1KB
- Open your virtual box.
- Create a new machine with following in mind:
 - Type: Linux
 - Version: Linux 64 bit other
 - Base Memory: 500 – 1024 MB
 - Processors: 2
- Do not create a new VM, rather choose “use an existing hard disk”
- Navigate to your owasp folder and select and create the VM
- Now the VM is there on your virtual box.

CSCS 495 A

- Select it and goto settings → Network settings and change network adapter from NAT to Bridge Adapter.
- Start the machine. You will get a command line interface.
- Type in username:password as root:owaspbwa
- You should get the prompt
root@owaspbwa:~#
- Type ifconfig and note down the IP.
- Open Firefox on kali machine and type the IP on URL bar.
- OWASP BWA page will be displayed.
- Finally check the communication of OWASP BWP with kali by pinging the ip of owasp bwa on kali terminal.

Task 2 [25]

In this task we will create a tryhackme.com account.

TryHackMe is an online platform designed to teach cybersecurity and ethical hacking through hands-on learning. It offers a wide range of interactive, gamified learning paths, challenges, and rooms that focus on practical skills related to penetration testing, cybersecurity, and network security. The platform is suitable for beginners as well as more advanced security enthusiasts, providing a variety of content to help users learn at their own pace.

Prepare a report using same format as for task-1.

Steps to follow:

- Goto googles and type tryhackme.com
- Click on the link provided by google.
- Once the page appears on screen, click on Join For Free button on top right.
- Complete the required information form and click sign up.
- Make sure you provide a correct email as they will validate it.
- Log into the portal and you see the dashboard page.
- Click on the Profile icon on top right side of the page, and from here click on Access.
- You will see a page saying Download configuration file.
- Click on it and a .ovpn file is downloaded with your user name.
- This is a configuration file used by open VPN. This file contains settings and parameters needed to establish a secure VPN connection between a user's device and a VPN server.
- While performing exercises in tryhackme environment, we may need to use machines to connect to tryhackme network. This can be done
 - Using OpenVPN (which is free)
 - Web based attack box (paid)

CSCS 495 A

- We will go for free.
- Once .ovpn file is downloaded, got kali and open a terminal.
- Navigate to your download directory. (You may copy the file on some other location of your liking)
- Type

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$ sudo openvpn raufbutt.ovpn
```
- This will configure settings and parameters required and finally you will see the message "Initialization sequence completed"
- Note that you will not see the prompt.
- If you need to use terminal now you should open a new tab.
- Open a new tab and type \$ ifconfig
- You will see a new interface "tun0" with an IPv4 address.
- This means we have now a connection
- Go back to the tryhackme page from where you downloaded the ovpn file.
- On right pane click on Refresh button and you will see a bunch of information:
 - VPN server name
 - Internal virtual IP address
 - Server status (which should say: online)
 - Connection (which should say: connected)
- Now let us try some exercise.
- Go to Learn → Search
- In search box type Linux
- Quite a few results will be displayed.
- Click on first room saying Linux Fundamentals Part 1
- Complete task 1 and task 2 (very straight forward)
- Go to task 3
- Click on green button saying Start machine
- You should see the target IP address on right pane and a linux machine on the left pane.
- Try complete the task using this machine.